

Islam Asous

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EDUCATION

Hunter College, City University of New York (CUNY)

Bachelor of Arts in Computer Science, Minor in Mathematics

New York, NY

expected Dec 2026

- **Relevant Coursework:** Data Structures & Algorithms, Programming Methods, Computer Theory, Computer Architecture, Database Management, Foundations & Practice of Modern AI Systems, Automation Engineering, Linear Algebra

EXPERIENCE

NYC Department of Transportation

Software Engineer Intern

Aug 2026 – Present

Remote

- Scoping and delivering a minimum viable product against a formal project scope and change-order process, translating stakeholder requirements into defined deliverables and timelines.
- Collecting and analyzing data to inform application requirements and track project progress.
- Communicating progress through written status reports and oral presentations to leadership, practicing stakeholder management and technical communication.

Project Sisyphus, Clemson University

Software Engineer REU

June 2026 – Present

Remote

- Maintain a authentication-tracking browser extension (*WXT, Vue 3, TypeScript, Dexie.js*) for NSF-funded research (\$650K, Award #2521037) on authentication-fatigue-driven password reuse across enterprise platforms.
- Built a real-time broadcast notification system (*Vue 3, Pinia, Supabase Realtime*) enabling admins to push instant announcements, enforcing role-based access via Postgres Row-Level Security.
- Embedded *Claude Code* into an agentic development workflow spanning implementation, testing, and cross-repository debugging (Supabase, OAuth, CI), accelerating feature delivery while maintaining 90%+ test coverage.

CUNY Tech Prep

Data Science Fellow

May 2026 – Present

New York, NY

- Selected for a competitive fellowship with students from across the 11 CUNY senior colleges where Fellows create end-to-end data science and AI projects using tools such as Python, SQL, Pandas, scikit-learn, and Hugging Face.
- Participate in weekly courses and learn industry best practices for exploratory data analysis (EDA), feature engineering, statistical analysis, data visualization, LLMs, RAGs, vector databases, and generative AI applications.

Hunter College Computer Science Department

Undergraduate Teaching Assistant

Aug 2025 – Present

New York, NY

- Supported 600+ students across Computer Architecture I & II, Computer Theory, and OOP/Data Structures & Algorithms by leading weekly review sessions and office hours.
- Co-designed quizzes, midterms, and finals for courses of 150–300+ students; reduced grading turnaround by building automated evaluation tools with faculty.

PROJECTS

AI Agent From Scratch, Tool-Calling Research Agent | *Python, LangChain, Anthropic Claude API, Pydantic*

- Engineered a tool-calling research agent using LangChain and the Claude API that autonomously decides whether to invoke a live web-search tool or answer directly based on the query.
- Eliminated parser-compatibility failures across 100% of tool-invoking and direct-answer queries by diagnosing a prompt-adherence bug and building a Pydantic-validated structured-output schema (topic, summary, sources, tools used).

RAG From Scratch, Local RAG Pipeline | *Python, Ollama, Llama 3.2, BGE Embeddings, Streamlit*

- Implemented a retrieval-augmented generation pipeline from first principles, implementing custom embedding storage, cosine similarity search, and prompt construction without relying on frameworks like LangChain.
- Improved answer grounding by iterating on prompt design against an in-memory vector database (locally-hosted BGE embeddings), surfaced through a Streamlit interface with an inspectable retrieved-context view.

For the Culture | *React, Vite, Express.js, Google Generative AI, Firebase*

- Expanded access to real-time food, health, and transit resources across all 5 boroughs in 8 languages by co-building and deploying a full-stack multilingual platform with a 4-person team, integrating a generative AI model for on-demand translation.
- Owned the React/Vite frontend architecture, including an interactive map consuming live NYC Open Data REST endpoints with real-time filtering by borough and resource type.

TECHNICAL SKILLS

Programming: C++, Python3, HTML/CSS, TypeScript, JavaScript, Rust, Swift/SwiftUI, Dart, SQL (PostgreSQL)

Frameworks, Cloud & Tools: Git/GitHub, React, Node.js, Express.js, Flutter, Firebase, Supabase, Docker, REST APIs

Certifications: AWS Skill Builder - Developer, Cloud Operator (SysOps), DevOps (In Progress); CodePath AI201: Applications of AI Engineering (In Progress); Databricks Generative AI Fundamentals (In Progress)